Brandon Chen and Shaunak Batra
Valuation for Financial Engineering Mini-Project 4

Future Hedging Strategy Discussed in Class

The hedging strategy presented in the slides is designed for a corn farmer (Farmer Brown) who expects to harvest 1,000,000 bushels of corn in July and has $2,000,000 in prepaid fixed costs. To protect against price drops and lock in a profit, the strategy in class uses the following parameters:

- **Position:** Short
- **Contract Month:** July futures
- **Exchange:** CME
- **Volume:** 200 contracts (at 5,000 bushels per contract, this perfectly matches the 1,000,000-bushel expected harvest)
- **Locked-in Price:** $4 per bushel
- **Financial Outcome:** By selling the entire expected yield forward at $4 per bushel, the farmer locks in $4 million in revenue, which guarantees a $2 million profit regardless of whether spot prices at harvest fall to $3 or rise to $5

Assumptions Made

- **Baseline, Costs & Optimization Objective:** The hedge is sized against a 1,000,000-bushel target harvest, and July futures start at $4.00 per bushel. We minimize the variance of cash sales plus hedge payoffs net of premiums and fees. We deduct $2 million in prepaid fixed costs and $0.20 per bushel in variable costs when reporting profit.
- **Distributions & Volumetric Risk:** Drought occurs with 15% probability. Yield is normally distributed with a mean and standard deviation of 1,080,000 and 120,000 bushels in normal weather, and 550,000 and 150,000 bushels in drought. Simulated yields are floored at zero.
- **Price Dynamics, Jump Risk & Consequences:** Terminal futures prices are normally distributed and floored at zero, with mean and standard deviation of $3.78 and $0.80 in normal weather, and $5.25 and $1.11 in drought. An independent pre-harvest spike occurs in 5% of scenarios and is drawn uniformly from $8 to $15. Without a spike, the interim price is drawn uniformly from $4.00 to $4.50. The modeled peak is the greater of that draw and the terminal futures price.
- **Basis Risk:** Local cash price equals terminal futures price plus an independent normal basis with mean −$0.15 and standard deviation $0.20 per bushel.
- **Margining & Liquidity Constraints:** We assume $500,000 is available for variation margin after initial margin and put premiums. If the modeled requirement exceeds that

amount, we close the CME futures position at the simulated peak. Any puts remain in place.

- **OTC Credit Line & Options:** We assume a $550,000 OTC variation-margin line, a flat OTC fee of $0.03 per bushel hedged, and a $0.33 per bushel premium for the $4 put.
- **Grid Search Boundaries:** We test short futures and long puts from 0% to 100% of target harvest in 5% increments, with a combined cap of 120%.

Performing the Analysis

- We wrote some Python code to perform the analysis, linked here.
- We simulated 10,000 scenarios and ran separate grid searches for CME shorts plus puts and OTC shorts plus puts. We excluded OTC positions that exceeded the credit line in any simulated scenario. We ranked each mix by the standard deviation of hedged sales proceeds and reported profit after production costs.

Output of Analysis

```
--- FINAL STRATEGY COMPARISON ---
                        Strategy Proceeds Std Dev (Target) Expected Profit 5th Percentile Profit
               Unhedged Baseline               $1,075,964      $1,553,768            $-147,177
                100% CME Futures               $1,999,682      $1,190,007          $-2,540,982
                  100% Puts Only                 $887,546      $1,610,369              $-7,923
Best CME Mix (5% CME + 100% Puts)                $886,076      $1,607,260             $-43,239
Best OTC Mix (5% OTC + 100% Puts)                $884,899      $1,609,303             $-37,700

100% CME futures liquidation rate: 30.1%
```

Recommended Strategy

- Farmer Brown should short July corn futures through OTC on 50,000 bushels (5% of the target harvest) and buy $4 July corn puts covering 1 million bushels. This mix produced the lowest sales-proceeds standard deviation among the strategies tested, at $884,899. The puts pay when July futures fall below $4, while the actual harvest is sold at the local cash price. The OTC credit line covers the short position's variation margin in every simulated scenario.

Recommended Strategy vs Expected Strategy

- We expected a puts-only hedge because a short futures position loses value when drought pushes futures prices higher. However, in the simulation, adding a small OTC short acts as a highly efficient cost-optimization tool. Because OTC futures have a much lower upfront fee ($0.03) compared to the expensive put options premium ($0.33), layering in a small low-fee forward slightly reduces the relative cost drag on the portfolio. This tightened the proceeds standard deviation from $887,546 to $884,899. Put spending

stayed the same, and the added short lowered expected profit by about $1,065, but we chose this mix because it successfully minimized the variance.

Recommended Strategy vs. In-class Strategy

- In class, Farmer Brown shorted 200 July CME futures at $4 per bushel, covering the entire 1 million-bushel harvest. With fixed production and zero basis, that hedge locked in $4 million in sales proceeds. Our strategy shorts only 50,000 bushels through OTC futures and uses $4 puts to protect against falling futures prices on 1 million bushels. The smaller short limits futures losses when drought reduces production and drives prices higher. Under our simulated margin assumptions, the full CME futures position was closed out in 30.1% of scenarios, and its proceeds standard deviation was $2.00 million. The recommended OTC mix reduced that figure to $884,899, slightly below the $886,076 achieved by the best CME mix.